Performing Multiple Linear Regression to Predict E-Commerce Customer Order Values

A Data Analysis of Retail Sales by:

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Who am I?

I am Nolan Mellot, a Data Analytics major who has a specialty in Data Engineering who is finishing their Master’s program.



Research Question:

Can a multiple linear regression model predict an e-commerce customer's order value using Sewell's Dataset (Sewell, 2026)?

Hypothesis:

Null Hypothesis

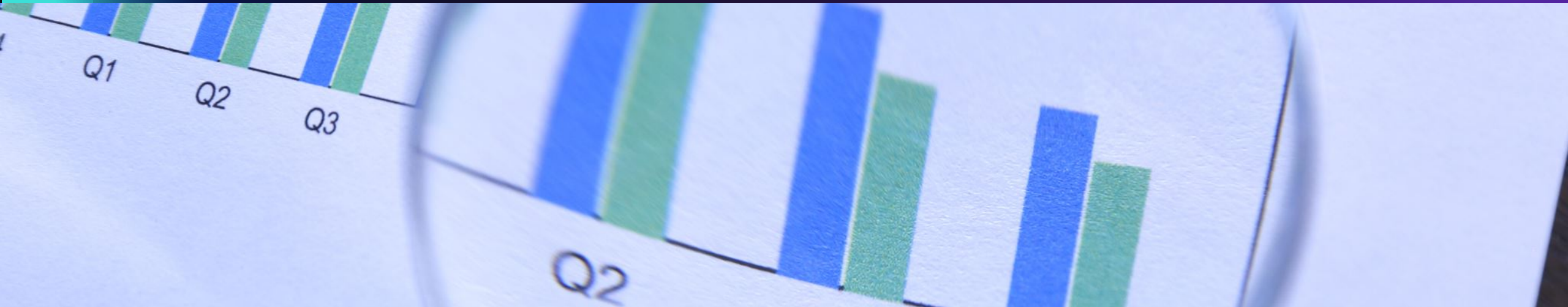
The combined set of primary drivers **does not** predict order price better than random chance.
Criteria: $\beta_1 = \beta_2 = \ldots = \beta_p = 0$ (All coefficients are equal to 0) or accuracy < 66%.

Alternative Hypothesis

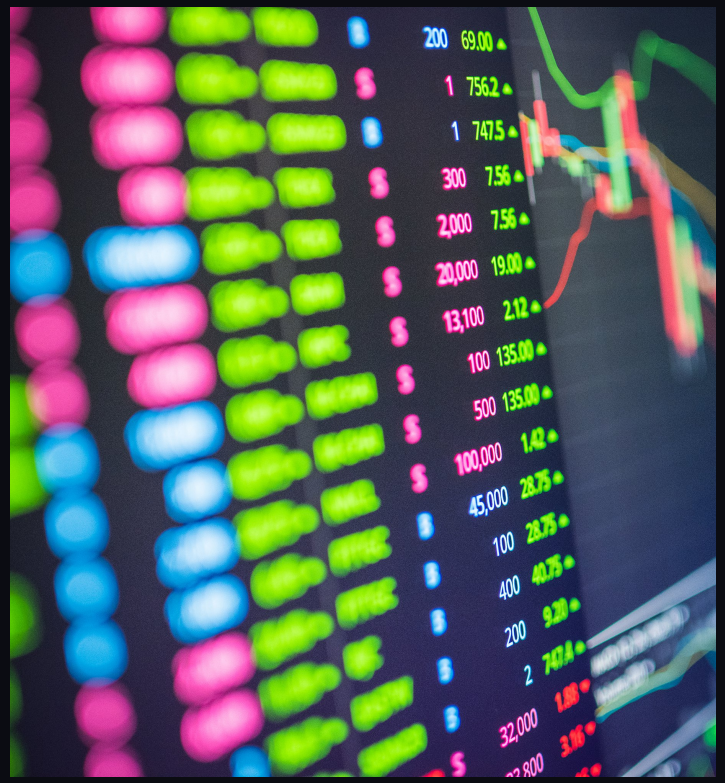
The combined set of primary drivers can predict order price better than random chance.
Criteria: At least one coefficient $\beta_j \neq 0$ (At least one coefficient is not equal to 0) and accuracy >= 66%.

The Beginnings of the Data Analysis Process

1. The data analysis started with data import and basic EDA (Exploratory Data Analysis) to examine the dataset.
2. Once the data was examined the unneeded data was removed.
3. Basic univariate and bivariate visualizations of the variables were created to visually inspect the data.



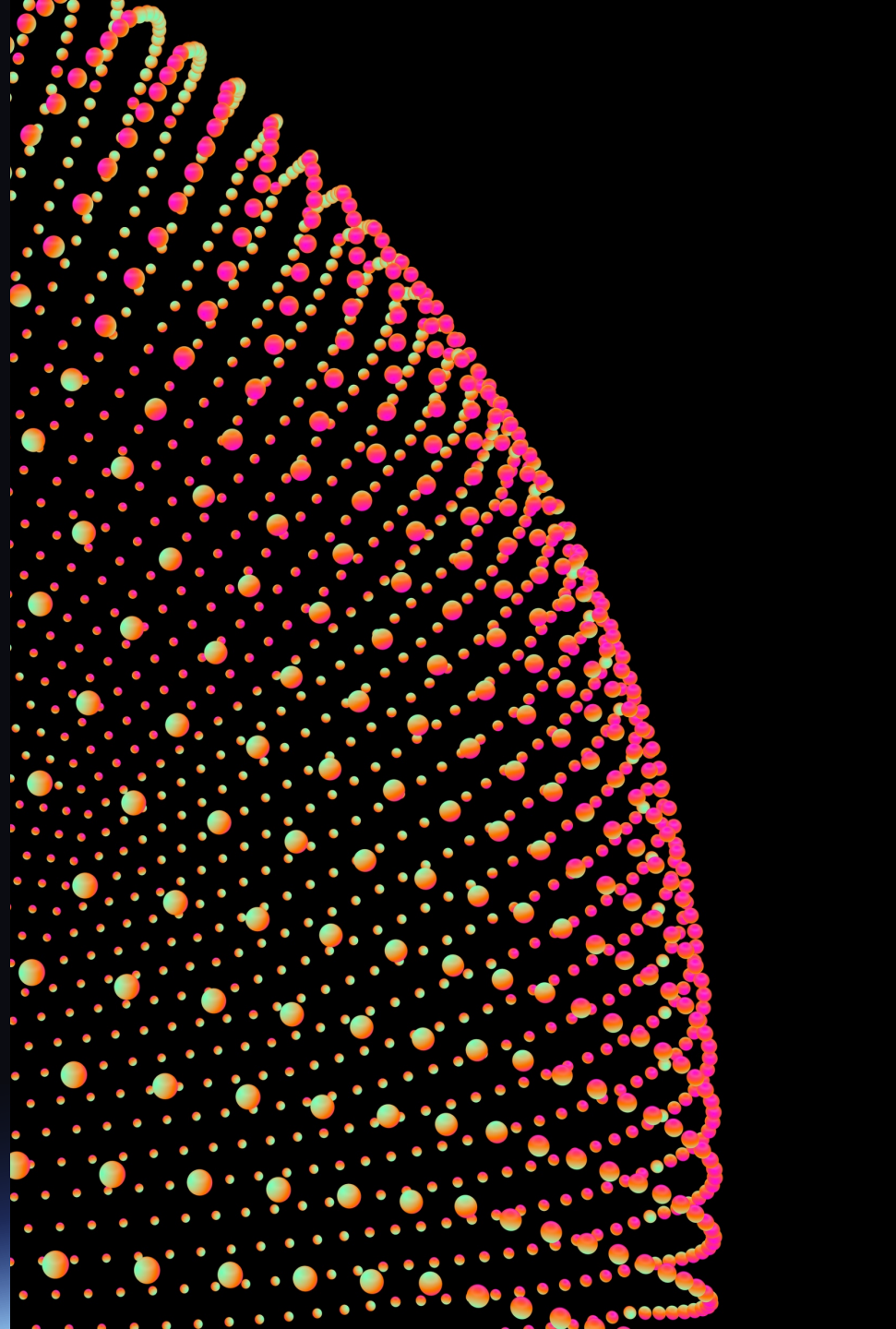
Preparing Variables for Regression



1. During visual inspection of the data, it was noted that many numeric variables had top-end outliers, these would later be removed using winsorization. Winsorization is a technique to preserve datapoints when outliers are present by capping the values (Couto da Silva, 2025).
2. For categorical variables, dummy variables were created as multiple linear regression can only use variables with numeric values.
3. The data was then split into training and test sets to allow for model evaluation later.
4. The training data was then scaled so that all variables would have equal weighting; to do this the values were scaled to their z-score, meaning the number of standard deviations away from center the value was, meaning most values will fall between -3 and 3.

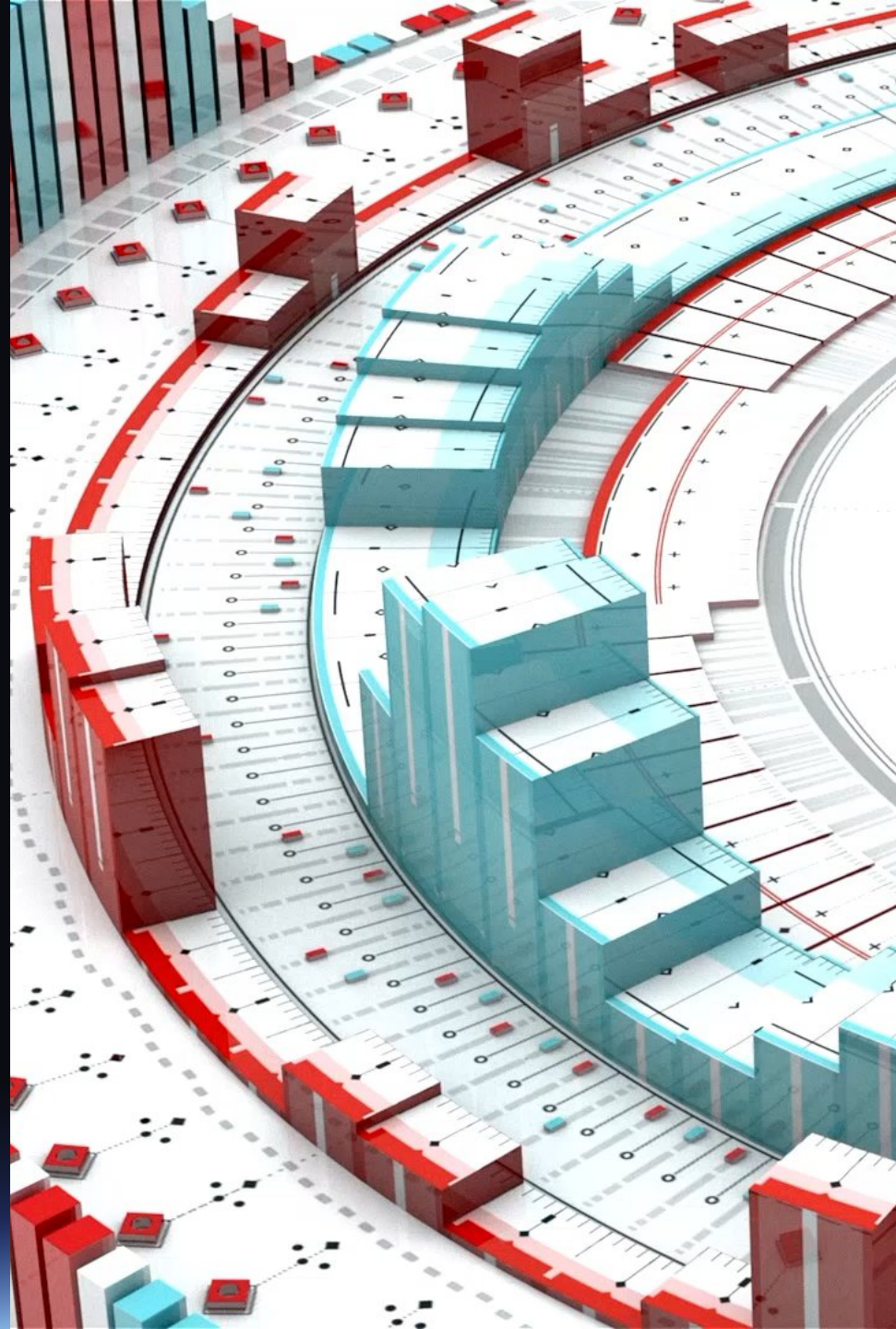
Regression and Model Refinement

1. The initial model was then created using all variables that were not removed during data preprocessing.
2. The model returned a low r^2 score meaning that the model was highly inaccurate.
3. The number of variables was then refined using forward stepwise elimination, a technique that adds the most statistically significant variable to the model until the p-value of the variables are above a set threshold.
4. The process of forward stepwise elimination left three variables; however, one of the variables was a dummy variable for the region variable.
5. As this was not the only dummy variable for the initial region variable, I needed to add the rest of the dummy variables for the initial region variable.
6. After the rest of the dummy variables were re-introduced to the model, a final model was created.



Model Prediction and Statistics

1. Before final predictions, all transformations done to the training set needed to be performed on the test set, this included scaling, and variable selection. The final trained model was then used to create predictions using the test set. Then the following model statistics were calculated:
    1. Root Mean Square Error (RMSE)
        1. Train: 24.6166
        2. Test: 24.9194
    2. Mean Absolute Error (MAE)
        1. Train: 20.1197
        2. Test: 20.2808
    3. r^2 score
        1. Train: 0.0351
        2. Test: 0.0209
    4. Adjusted r^2 score
        1. Train: 0.0023
        2. Test: 0.0023



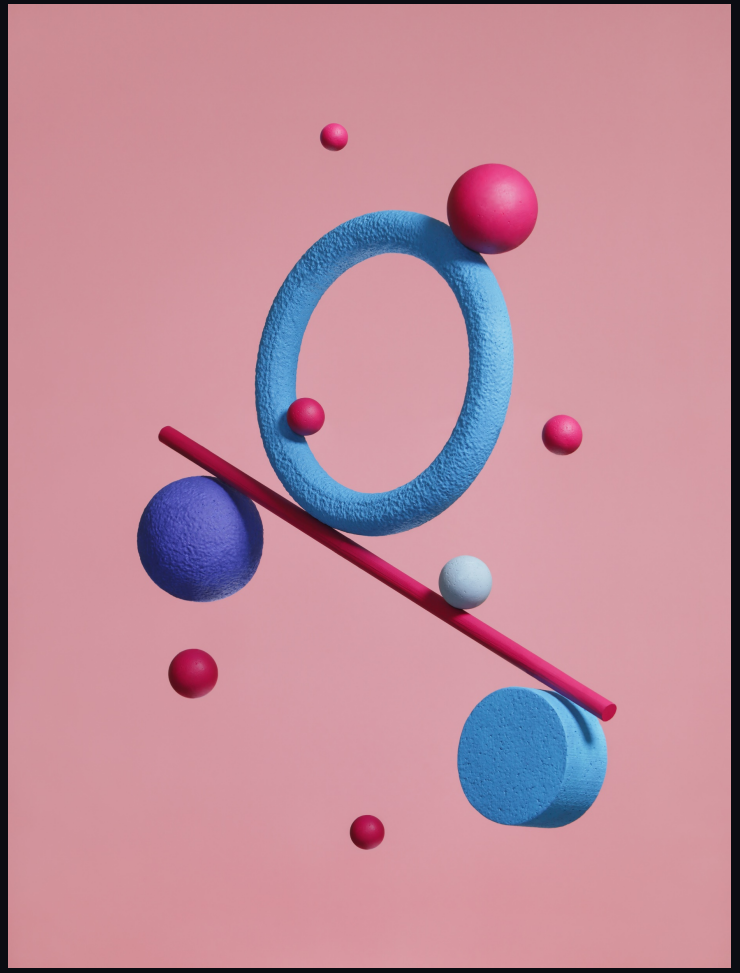
What Do the Statistics Mean?

RMSE: From the RMSE of both the training and test set, we can see that there are large errors, without much difference between the training and test set. Although there are large errors, we can see that the errors are consistent, meaning that there is little to no under-or overfitting.

MAE: The same information from the RMSE can be applied here due to the similarity in values between these statistics. The MAE of both sets also proves the lack of presence of under-or overfitting.

r^2 score: The r^2 score tells use the accuracy of the model, meaning what percentage of variance is captured in the model. As this value was very low for both sets, there is little variance captured in the model in general.

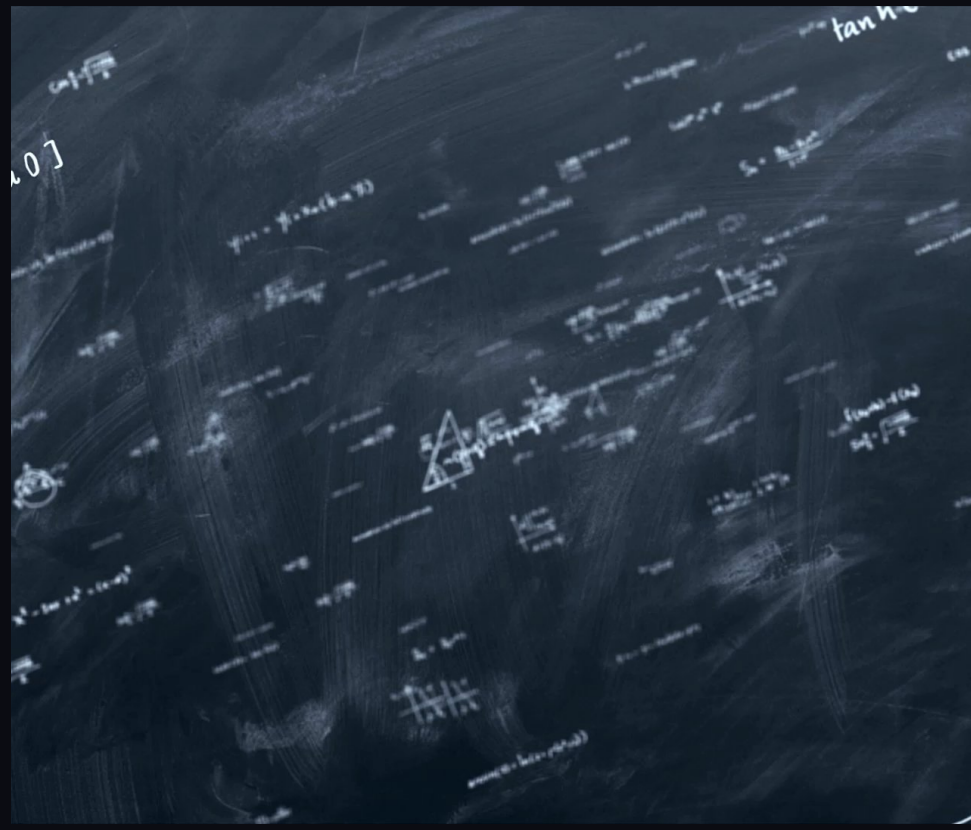
Adjusted r^2 score: The difference in this statistic, to the last, is that it punishes having non-statistically significant variables included in the model. There is little difference between the different r^2 scores as the model is so inaccurate.



The Regression Equation

Order Value = 45.56 +
3.95('z-score income_k') + 2.01('z-score prior_3mo_visits') +
0.25('z-score region_South’) - 0.64('z-score region_Northeast’) - 1.87('z-score region_West’)

As seen above: the constant is a very large value compared to the coefficients, keep in mind that z-scores typically only range from -3 to 3. Another thing to keep in mind is that as the region prefix variables are categorical and rely on one another, at most, one can be true.





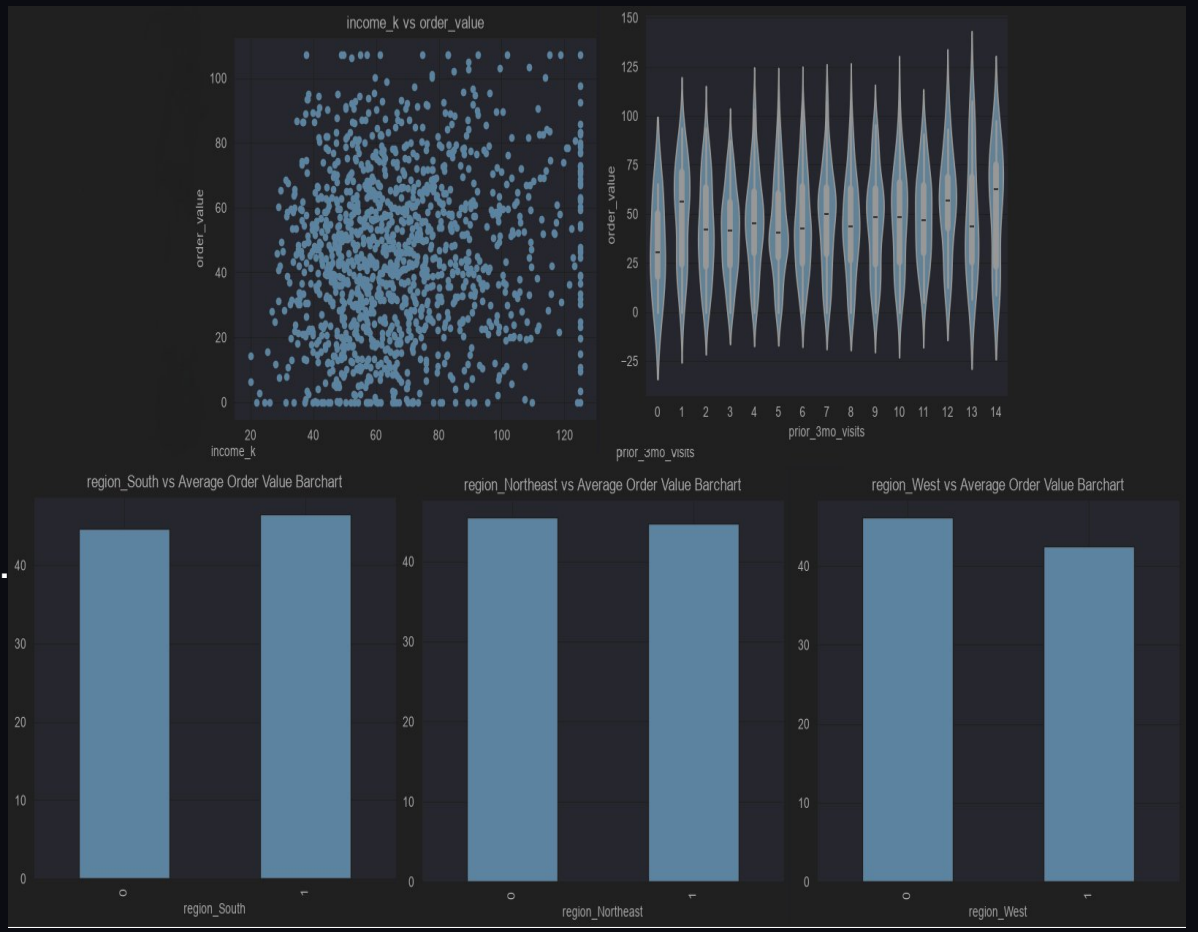
The Five Assumptions of Multiple Linear Regression

The Five Assumptions of MLR according to Bobbit (Bobbit, 2021)

1. Linearity – Is there a linear relationship between the independent variables and the dependent variable?
2. No Multicollinearity - Do the independent variables have high correlation with one another?
3. Independence – Are the independent variables independent of one another?
4. Homoscedasticity – Is the variance in residual errors uniform across all levels of the independent variables?
5. Multivariate Normality – Is the variance in residual errors normally distributed?

Linearity Test

Linearity is tested through inspection of bivariate visualization between each independent variable and the dependent variable. From the visualizations on the right we can see that there is a rough linear relationship between the variables and the dependent variable. In the categorical variables, we can see that there is a noticeable difference between average values. From these visualizations, there is enough of a linear relationship to say that this assumption is true.



No Multicollinearity Test

The test for multicollinearity is simple as it can be imported as a function. The statistic used to test for this is Variance Inflation Factor (VIF). The closer to 1 the value is, the lower the multicollinearity of that variable. Values under 5 can be considered low enough to prove no multicollinearity (Saluja, 2024). As seen in the table to the right, all values are under 2, meaning that there is little to no multicollinearity, meaning the assumption of no multicollinearity can be stated as true.

	VIF	variable
0	1.003602	income_k
1	1.003473	prior_3mo_visits
2	1.757323	region_South
3	1.608684	region_Northeast
4	1.624815	region_West

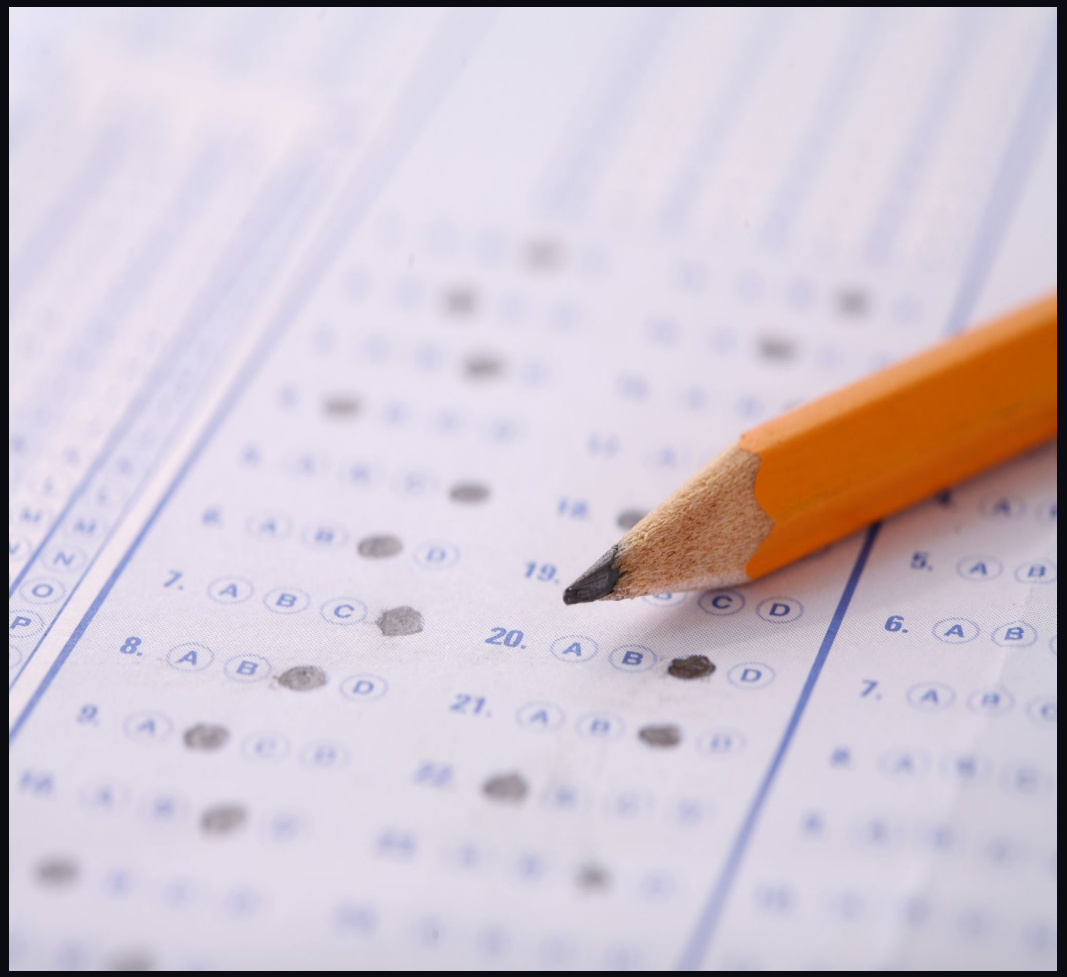
Independence Test

The test for Independence is simple as it can be imported as a function. The test used is a Durbin-Watson test and it finds the amount of autocorrelation in the model. Values between 1.5 and 2.5 can be seen as little to no autocorrelation. The value for the model is around 2.0368, meaning there is very little autocorrelation, meaning there is independence. The assumption of independence can be stated as true for the model.



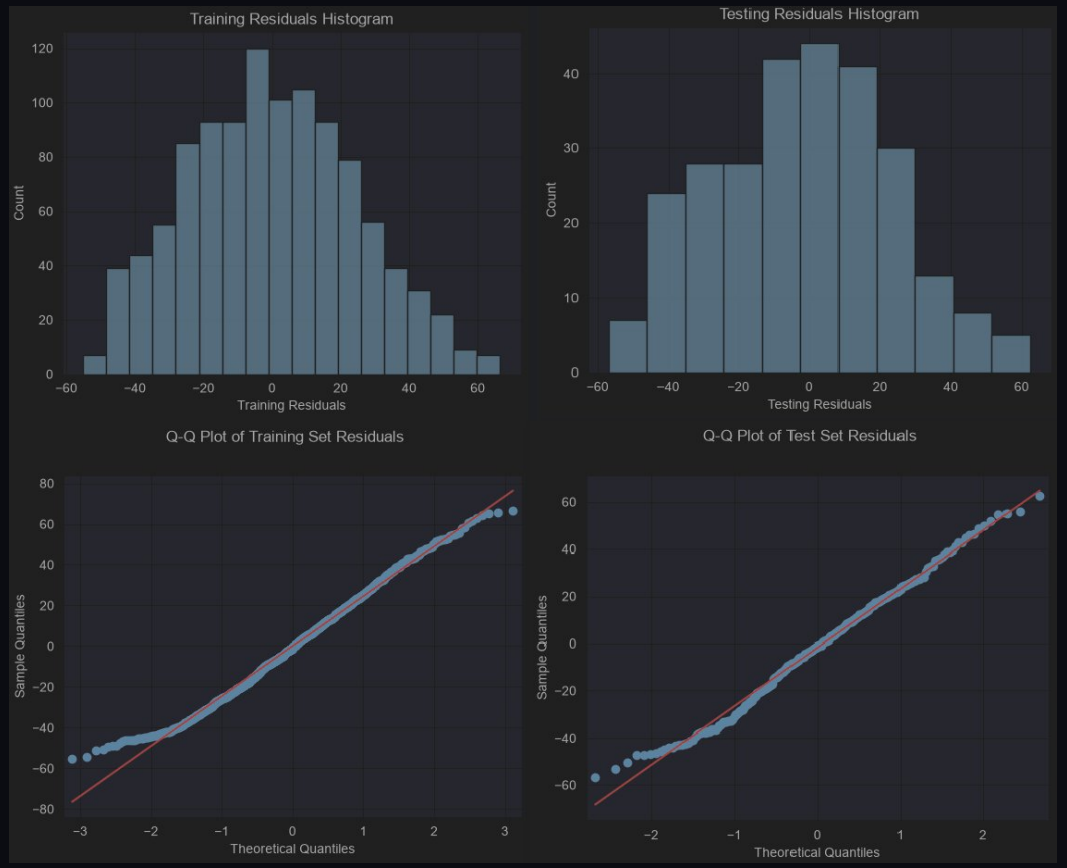
Homoscedasticity Test

To test for homoscedasticity, there must be a lack of evidence to support heteroscedasticity. The test used to check for heteroscedasticity is a Breusch-Pagan Test. When running the test, we get a p-value of .4544, meaning there is a lack of statistical evidence of heteroscedasticity, meaning there is homoscedasticity. From this we can say that the assumption of homoscedasticity is true.



Multivariate Normality

The test for multivariate normality is a visual test using visualizations utilizing model residuals. The histograms are used to check for normal distribution, a bell curve, and the Q-Q plots are to see how closely the plotted points follow the red center line. From the histograms we can see that there is slight right skewedness, from the Q-Q plots we can see that the points roughly follow the red line with deviation at each end, which is normal for normality, as this is a sign of under dispersion in the data (Waples, 2024). The plots used for this test are shown on the right. From the inspection of these visualizations, we can say that the assumption of multivariate normality is true.



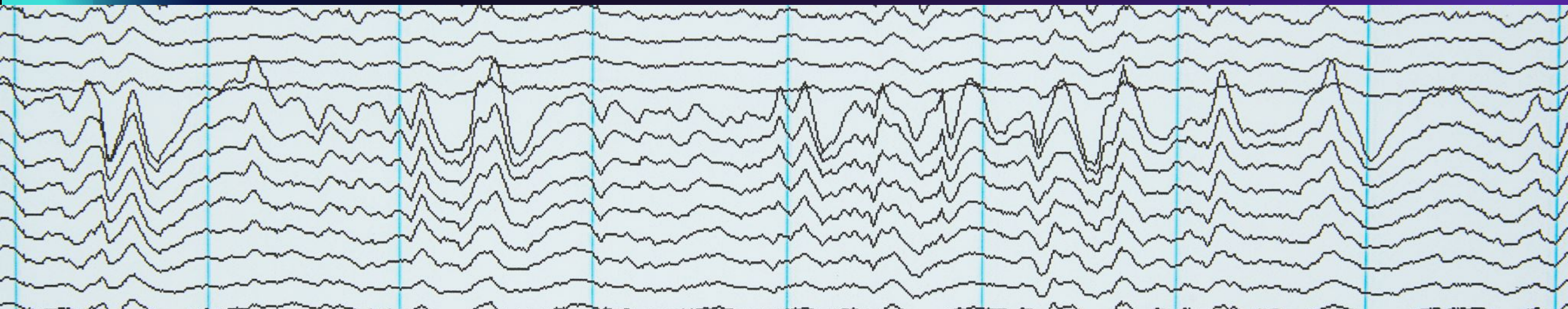
Why Check the Assumptions of Multiple Linear Regression?

Although the model is proven to be inaccurate and cannot predict order values, checking the assumptions of MLR is important for reliable results. Ensuring the results of the regression are reliable ensures that the answer derived from the equation, there is little explained variance, is in fact true. If the model did not meet the assumptions, then the results could be seen as unreliable, and could have missed potential explained variance in the data.



Limitations of Study

Multiple linear regression can only explain linear correlation between dependent and independent variables. I only chose to predict order values that were not equal to 0, where an order takes place.



Proposed Actions

1. Continue to attempt to predict order value using newly collected data with new variables.
2. Attempt to predict a different variable, such as using logistic regression to predict whether a customer will make a purchase.



Conclusions



Although a regression equation that can accurately predict customer order value was not created there is various information learned during this study.

We can see that the variables in the current dataset do not explain much variance in customer order value, meaning decisions around customer order value should not include the variables included in the study.

As the assumptions of multiple linear regression were tested, we can assume that the information gained from this study is reliable and can be used in a market setting.

Sources:

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